ARTICLE IN PRESS

Journal of King Saud University – Computer and Information Sciences xxx (xxxx) xxx

Contents lists available at ScienceDirect

Journal of King Saud University – Computer and Information Sciences

journal homepage: www.sciencedirect.com

A new hybrid deep learning model for human action recognition

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ARTICLE INFO

Article history:
Received 9 January 2019
Revised 4 September 2019
Accepted 8 September 2019
Available online xxxx

Keywords:
Deep learning
Recurrent Neural Networks
Gated Recurrent Unit
Video classification
Motion detection

ABSTRACT

Human behavior has been always an important factor in social communication. The human activity and action recognition are all clues that facilitate the analysis of human behavior. Human action recognition is an important challenge in a variety of application including human-computer interaction and intelligent video surveillance to enhance security in different domains. The evaluation algorithm relies on the proper extraction and the learning data. The success of the deep learning led to many imposing results in several contexts that include neural network. Here the emergence of Gated Recurrent Neural Networks with increased computation powers is being adopted for sequential data and video classification. However, to have an efficient classifier for assigning the class label, it is very necessary to have a strong features vector. Features are the most important information in each data. Indeed, features extraction can influence on the performance of the algorithm and the computation complexity. This paper proposes a novel approach for human action recognition based on hybrid deep learning model. The proposed approach is evaluated on the challenging UCF Sports, UCF101 and KTH datasets. An average of 96.3% is obtained when we have tested on KTH dataset.
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1. Introduction

In the last few years, there has been a growing interest of human actions recognition (Yeffet and Wolf, 2009; Shu et al., 2014; de Oliveira Silva et al., 2017) in real-world environment, which finds applications in a variety of domains. Vera et al. (2011) developed an intelligent human-computer interaction application that talks with an audience using augmented mirror. This application presents a virtual character based on human motion capture and detects head movements. In the same domain, Lopatovska and Arapakis (2011) proposed an application to analyze a human expression and illustrated their role in human information behaviour. Moreover, the increasing crime rates and need of surveillance systems encouraged researchers to develop the intelligences surveillance applications. Ko (2011) presented in his paper a visual surveillance application to detect abnormal human behavior implemented on Performance Evaluation of Tracking and Surveillance (PETS) datasets. Gowsikhaa et al. (2012) published a survey titled "Automated human behavior analysis from surveillance videos" in which, he explained the role of human activity analysis, human motion and human tracking in surveillance systems (Dey et al., 2016). In the other context, Choi et al. (2013) applied the deep belief network algorithm to model and predict human behavior in smart homes for increasing the quality of inhabitant's life. Recently, there has been rapidly growing of internet shopping. Yi Jin et al. (2016) presented a study on the relationship between purchase intention, online shopping behavior, perceived usefulness and subjective norm.

Precise actions recognition (Pulgarin-Giraldo et al., 2017) is an extremely challenging task due to change in viewpoint variations and clutter backgrounds. Therefore, we can mention, that the most popular state-of-the-art methods for human action recognition (Jian et al., 2015; Geng and Song, 2015a) uses engineered motion (Wang et al., 2011; Toshev and Szegedy, 2014)and texture descriptors calculated around spatio-temporal interest points (Geng and Song, 2015b). In addition, most of these approaches follow the conventional paradigm of pattern recognition. These approaches are based on two steps: evaluates complex handcrafted features from video frames and learns classifiers based on the obtained features. In real-world scenarios, it is infrequently known which features are

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Peer review under responsibility of King Saud University.

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https://doi.org/10.1016/j.jksuci.2019.09.004
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Please cite this article as: N. Jaouedi, N. Boujnah and M. S. Bouhlel, A new hybrid deep learning model for human action recognition, Journal of King Saud University – Computer and Information Sciences, https://doi.org/10.1016/j.jksuci.2019.09.004

very important for the task at hand, since the choice of features is highly problem-dependent. Particularly, for human action recognition, action classes may appear different in terms of their appearances and motion patterns (Kale et al., 2016).

The present paper explains the importance of features to classify and recognize human action. Yet, this condition is strongly related to the techniques and algorithms used to extract complex features in sequences of video. Moreover, In order to better classify human action, this paper develops a novel approach which attempts to use the motion tracking, by the human tracking, as well as the spatial features extraction in a video sequence. To have a strong feature vector for classification, two techniques were applied; Gaussian mixture model (GMM) and Kalman filter (KF) methods to detect and extract the moving person, and Gated Recurrent Neural Networks to collect the features in each frame and predict human action. In addition, to a semantically meaningful, such features should present better descriptions of human action. In this challenging, we evaluate our method on the UCF (University of Central Florida) Sports action, UCF101 and KTH human action dataset where we find variety of action in different background. The experimental results show that the proposed approach is a very competitive algorithm for the human action recognition problems. Also this paper concludes that the proposed method can be used in a variety of applications.

The remainder of this paper is organized as follows. The second section discusses the related work and the research issues which surveys the previous works of motion detection, object tracking and pattern Recognition. Third section presents the proposed hybrid deep learning model using Gated Recurrent Neural Network (GRNN), GMM and KF. In fourth section, experimental results and comparison with state of the art are explained.

2. Related works

Most studies on the recognition of human action are focused on human detection and motion tracking. Many approaches have been proposed for human detection using background subtraction. Gaussian model uses a statistical modeling of the pixels composing the frames of the video according to their intensity, their color or their texture. In the case where the videos have weak motions in the background, Wren et al. (1997) model each pixel of the background by a Gaussian distribution. Sigma-delta is another approach for background subtraction. This filter is proposed by McFarlane and Schofield (1995) to compute two orders of temporal statistics for each pixel in a video sequence providing a multiple field observation and decision-making at the pixel level. Kim et al. (2005) present an algorithm for the multimodal foreground/background segmentation. They construct for each pixel of the background a codebook composed of a several codewords. There are many approaches for human detection but none is significantly more successful than the others. Gaussian Mixture Model (GMM) (Stauffer and Grimson, 1999) is the most effective method in noisy backgrounds.

Moreover, the motion tracking is an important feature. Indeed, better motion exploitation is better action recognition. Therefore, once the object has been detected we can track its path. The tracking of moving objects can track and display the movement of objects. Such tracking can provide important data, as the speed and acceleration of an object. In addition, in recent years, there has been much work on algorithms of the tracking objects in a video sequence. Optical Flow (Zhang et al., 2017) and his improved, Histograms of Optical Flow (HOF) and Motion Boundary Histograms (MBH), were effective for detecting and tracking moving object. Wang and Schmid (2013) have illustrated the performance of HOF and MBH for the human body tracking using Hollywood2, Olympic Sports, Human Motion Database Benchmark HMDB51 and UCF50 datasets. Despite these advantages, Optical Flow is failed to track moving object if there is a change of lighting moderate and moving objects intensity. Kalman filtering (liu et al., 2007; Jang et al., 2002) presented a good performance in lighting changes. Indeed, it is very popular in the research field of navigation and aviation because of its magnificent estimation. Since then, the Kalman filter is an estimator for what is called the "linear quadratic problem". The linear quadratic problem focuses on the estimation of the instantaneous "state" of a linear dynamic system perturbed by white noise. In addition, the Kalman tracker adapts the learning parameters of the adaptive background module. Moreover, Kalman filter was successfully applied by Rameshbabu et al. (2012) to track human body while walking and detect her face. As well as Patel and Thakore (2013) used Kalman filter to track any single moving objects in indoor and in outdoor environments on surveillance video of Context Aware Vision using Image-based Active Recognition (CAVIAR) and PETS datasets.

Recently, deep learning became the most successful method. It is used to learn a hierarchy of features (Geng and Song, 2015a; Xu et al., 2010) in many applications including video (Shi et al., 2011; Idrees et al., 2016), image, speech and signal processing. Therefore, the feedforward neural network models can achieve state-of-the-art accuracy in object classification exceeding sometimes human-level performance. These models are trained by a large set of labeled data and neural network architectures that contain many layers. Baccouche et al. (2011) and Latah (2017), introduced a multiple 3d Deep Learning and 3D Convolutional Neural Networks for human action recognition. His techniques obtain a high performance on KTH dataset. In addition, the recent work of Deldjoo et al. (2017) presented a deep learning for movies recommendation with successful results. However, some drawbacks are found. Sharing parameters across the time is insufficient for capturing all of the correlations between input samples. Additionally, local connectivity limits the output in a function of a small number of neighboring inputs samples. To date, Recurrent Neural Networks (RNN) has demonstrated great success in sequence labeling and prediction tasks such as handwriting recognition and language modeling. Therefore, in several applications involving sequential or temporal data (Veeriah et al., 2015), various types of hidden units for RNN have been used to solve range of problems with impressive results. The Long Short-Term Memory unit (LSTM) proposed by Hochreiter and Schmidhuber (1997) is used with RNN in diversity tasks such as speech recognition (Graves and Jaitly, 2014; Chorowski et al., 2015), video captioning (Karpathy et al., 2014; Vinyals et al.,2015; Grushin et al.,2013) and handwriting recognition. Thus various architectures of LSTM networks are created to optimize a variety of applications. Bidirectional LSTM (BLSTM) networks have been proven by Alex Graves (Graves et al.,2009; Eyben et al.,2009)for framewise phoneme classification, also it is used to model a multi-stream framework (Wollmer et al., 2011) for continuous conversational speech recognition. LSTM Projected (LSTMP) have been proposed by Hasim Sak for Large Scale Acoustic Modeling (Sak et al., 2014). Cho et al. (2014) created, a simplify model of LSTM unit, the Gated Recurrent Unit (GRU). GRU is a lighter version of an LSTM where the complexity in the structure is reduced by decreasing the gates. Much previous works have justified that GRU incurs shorter run-time compared to LSTM. GRU has empirically demonstrated their ability to model long-term temporal dependency in various task on computer vision. Recently in (2017), Agarap have used a Neural Network architecture combining Gated Recurrent Unit and Support Vector Machine SVM (SVM) for intrusion detection in network traffic data.

The contribution of this paper focus on how can extract better features vector to best classify human action. The present work exploits the performance of the gating network signals of GRU to

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features extraction and video classification. Moreover, to improve the feature vector, two methods are used to determine the motion feature of person in sequence of video.

3. Proposed approach

3.1. Human motion tracking

In order to better exploit the performance of gated recurrent neural network in video classification, it is necessary to improve the features extraction methods. The goal of features extraction is to reduce the size of the data to be processed and the computation time. We have considered GMM and Kalman filter to around the moving person in each frame of video. Only data on the bounding boxes will be used by Gated Recurrent Neural Networks.

In order to detect moving persons in dynamic scene, background modeling is an important step. Gaussian mixture model is the most effective technique for human motion detection from background. GMM is a statistical model expressed according to a mixed density. It is generally used to parametrically estimate the distribution of various object features by modeling them as a sum of several Gaussians. A Gaussian is characterized by variance, mean, and amplitude. These parameters are optimized according to a maximum likelihood. This procedure is done iteratively via the expectation-maximization (EM) algorithm. For human detection all observations in each pixel are modeled by mixture of Gaussians. Indeed, GMM attributes a constant number of Gaussian distributions for each pixel in order to estimate the reference frame. If there is weak motion (no variation of pixel values) all Gaussian distributions approximate the same values, therefore one distribution is necessary. On the other case, if there is some motion and the pixel values are changes, it is necessary to determine the approximate number of Gaussians.

Kalman filter is a linear filtering algorithm that optimally estimates the state of a dynamic linear system given a set of measurements (or observations), when the observation noise is Gaussian additive. It has been extensively used for human tracking based on two cyclically phases, prediction and correction. In the prediction phase, the estimate of the current state is measured from the previous time step. This estimate is also called a priori state estimate. It does not require any measurement value. The Kalman filter estimates the position, velocity and acceleration of person in each frame of the video. It gives a precise result when there should be any variation in the speed of moving person. It can exploit for best results by taking the linear model and Gaussian noise. Moreover, Kalman Filter is a recursive estimator means that the estimation of the current state requires the previous state and its current measurements. These two are sufficient to estimate the current state. The Kalman filter averages the system state prediction with a new measurement using a weighted average phenomenon to obtain more accurate estimated values. The main purpose of using GMM and Kalman method is to reduce computation time and complexity by the bounding boxes of moving person in each frame of video.

3.2. Recurrent Neural Networks model for action classification

To date, a study of Recurrent Neural Networks with different types of hidden units have been illustrated their performance in diverse applications of video classification and action recognition. In order to reduce the variables number of RNN and the parameters of hidden unit, we integrate in each node of standard RNN a memory cell of Gated Recurrent Unit. This Unit is used to model the dependency temporal long-term and solves the problems of vanishing gradients. For video classification and human action recognition, the GRU simplifies the Gated Recurrent Neural Network on two gates, a reset gate and an update gate. The reset gate determines how to combine the new input frame with the previous memory, and the update gate control what to keep from the previous memory. Indeed, GRUs have been successfully used for sequential data structures. The prediction of the activity label for an action depends on the actions registered before. The recurrent memory management of GRUs allows us to model the problem considering those sequential dependencies. In additions, GRNN networks are particularly well suited for modeling problems in which temporal relations are relevant and the time gaps between the events are of unknown size. While the GRUs are the central elements of the proposed model of human action prediction, they can be organized on three parts in the Fig. 1: The inputs data, the GRU sequence modeling and the prediction module.

The inputs data receive the results of GMM and Kalman Filter which are the succession of frames, with small size, that present only moving person. The size of input data differs from one video to another. Indeed, the size depends on the distance between the camera and the person. These inputs are then fed to the GRNN hidden layer to extract the relevant features. Then, the prediction module is used to predict action. The prediction of human action is based on two major phases: training and test. The training phase is presented in the step 1 and 2, and the test phase is presented in step 3.

Training the Gated Recurrent Neural Networks model:

- Input the smalls video frames presented only the moving person
- Initialize the learning parameters weights and biases with arbitrary values (they will be adjusted through training).
- Compute the cell states of GRU using the input frames X_i and its learning parameters values.
- Adjust the weights by an optimization function. The weights are modified using the Back propagation through time, to minimize error and loss function. The adjustment is performed by the following equations.

$$W^{t+1} = W^t - \gamma \frac{\partial E}{\partial W^t} \tag{1}$$

$$U^{t+1} = U^t - \gamma \frac{\partial E}{\partial U^t} \tag{2}$$

where γ present the learning rate, $\frac{\partial E}{\partial W^t}$ and $\frac{\partial E}{\partial U^t}$ are the gradient values and E is the error and loss function.

- Save the trained GRNN model.

Test: We compile the model obtained using a different video. The outputs of our proposed model are the distribution of a probability of each class, where each class presents one human action. Finally, the test video belongs on class which has a max of probability.

4. Experiments

4.1. Dataset and experimental setup

We performed experiments on the UCF Sports Action, UCF101 and KTH datasets to demonstrate the important characteristics of our model for action recognition.

UCF Sports[1]: UCF Sports is one of the earliest actions recognition dataset. It consists of a set of actions collected from various sports.

[1] http://crcv.ucf.edu/data/UCF_Sports_Action.php.

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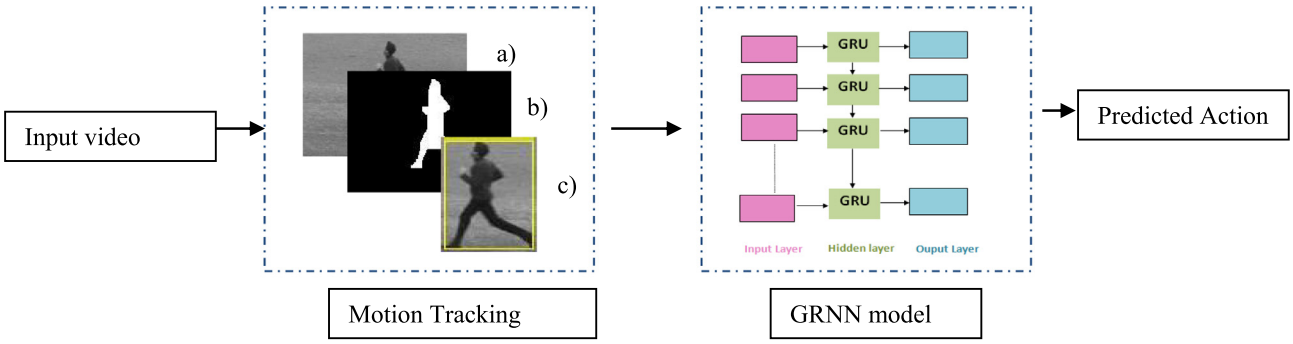

Fig. 1. Schematic overview of human action prediction model: In the motion tracking we are used GMM and Kalman Filter: a) present the frame of running person in KTH dataset b) the human motion detection based on background subtraction method using GMM. c) The bounding boxes of human motion based on tracking method using Kalman Filter. The GRNN model divided into three layers (input layer, one hidden layer and output layer).

The dataset consists of 150 videos, with the resolution of 720 × 480, for 10 human actions. The frames number for this dataset varies from one video to another. It is from 30 to 130 frames maximum in each video.

UCF101[2]: UCF101 (Soomro et al., 2012) is the largest and the one of the most challenging action dataset in terms of actions and scale. It contains a variety of action with a large variations in camera motion and cluttered background. The dataset consists of 13,320 video, of maximum 150 frames per video with the resolution of 320 × 240, for 101 human actions divided into five types, Human-Object Interaction, Human-Human Interaction, Playing Musical Instruments, Body-Motion Only and Sports. To evaluate the proposed method two types are used body motion only and sports with 66 actions.

KTH[3]: The KTH is the most popularly used public dataset for human action. It contains 6 types of video actions and resolution of 160 × 120 (walking, jogging, running, boxing, hand-waving and handclapping). The frames number used for video classification is around 100 frames per video (Table 1).

4.2. Results and discussion

In order to validate our approach, the experimental results are conducted on a recent version of Matlab codes on a laptop computer with Intel Core i7-8550U 8th Generation Processor 4.0 GHz and 8 GB memory. The dataset was split, for video classification, into Three-quarters of videos for the training and one-quarter for the test.

The proposed model divided into four major steps: Firstly, we implemented the GMM and Kalman filter to extract human motion in each frame of KTH dataset. These methods are successfully used and the results are presented in our paper (Jaouedi et al., 2016a). Secondly, KNN method is used with GMM and Kalman Filter for human action classification in which the classification rate achieved 71% in Jaouedi et al. (2016b). Thirdly, The Gated Recurrent Neural Network model was implemented. To do the training of our model, we have used 100 frames in the inputs, 32 GRU hidden nodes, 128 batch size, 30epochs, categorical cross-entropy as the loss function and Stochastic Gradient Descent (Hochreiter and Schmidhuber, 1997) as the optimizer. With Softmax activation function in the output layers, the accuracy of video classification was around 93% using KTH dataset.

Fourthly, to more improve the classification rate, around of 2000 videos are used from three datasets: 150 videos from KTH dataset, 150 videos from UCF Sports and 1700 videos from UCF101 dataset. Indeed, we have used the hybrid deep learning with motion tracking. The human tracking of each frame in video sequence are extracted by GMM and KF then the output of these methods are used by our GRNN model to extract feature and predict human action. For action classification, we have taken a test video as input in our hypride GRNN model trained. The Softmax method is used to determine the accuracy and loss in the training and the validation steps. The output of our approach is the ranking of K class of each dataset used in input.

Table 1
Experimental Setup.

Parameters	Values
The maximum number of frames in each input video	n = 100
The resize of input frames for motion tracking	224 × 224 × 3
Learning rate	0.1
The batch_size	128
The number of GRU hidden nodes	g = 32
Maximum epochs	30

The measurement results of classification rates is used 10 actions of UCF Sports dataset (nbclass = K = 10), 66 actions of UCF101dataset (nbclass = K = 66) and 6 actions of KTH (nbclass = K = 6). Indeed the validation loss decreased on 1% for UCF Sport (Fig. 3) and 0.37% for KTH dataset (Fig. 2). Moreover for the UCF 101 the validation accuracy achieved 97% in some actions (Fig. 4).

In the experimental results we have used some selected YouTube videos of flowers and sea to treat exceptions and to put the system into operation against false or fake human actions. Our proposed method added a new class in the three dataset (Example K = 7 for KTH dataset) which the seventh class contain the no action or the false human actions. The classification time of the proposed method do not exceed 40 min for both training and testing that is much more better than Jian et al. (2015) that in their method the training and testing time of the proposed CNN model is around 1 h and 8 min.

4.3. Comparison with state of the art

Table 2 shows the performance comparison with other approaches on the same KTH, UCF Sport and UCF 101 Human actions dataset. Indeed our approach, based on hybride deep learning, achieved an average 96. 30%. Overall, our method gives comparable results with the best related work on KTH dataset: de Oliveira Silva et al. (2017) are used the Global Silhouette and Local Optical Flow for human action recognition. The proposed model achieved an accuracy rate 95% and Latah (2017) exploit the performance of 3D convolutional neural networks model with the Support

[2] http://crcv.ucf.edu/data/UCF101.php.
[3] http://www.nada.kth.se/cvap/actions/.

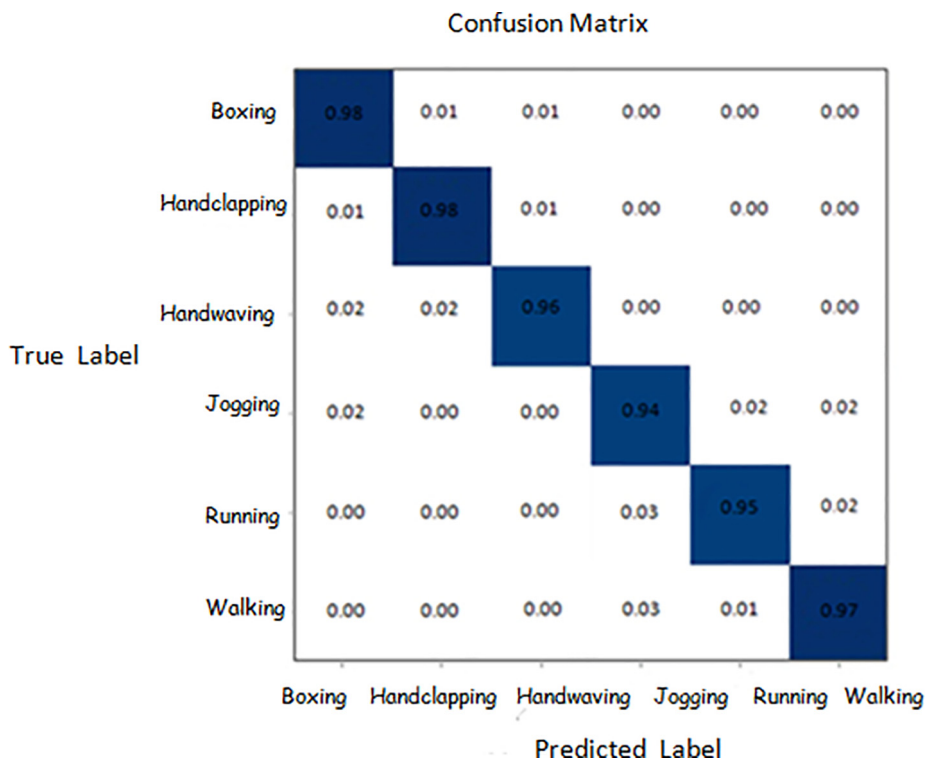

Fig. 2. The confusion matrix presentation of six actions of KTH dataset: Our model provides a high performance in all the actions. The validation accuracy varies between 98% for boxing and handclapping actions, and 94% for handwaving action.

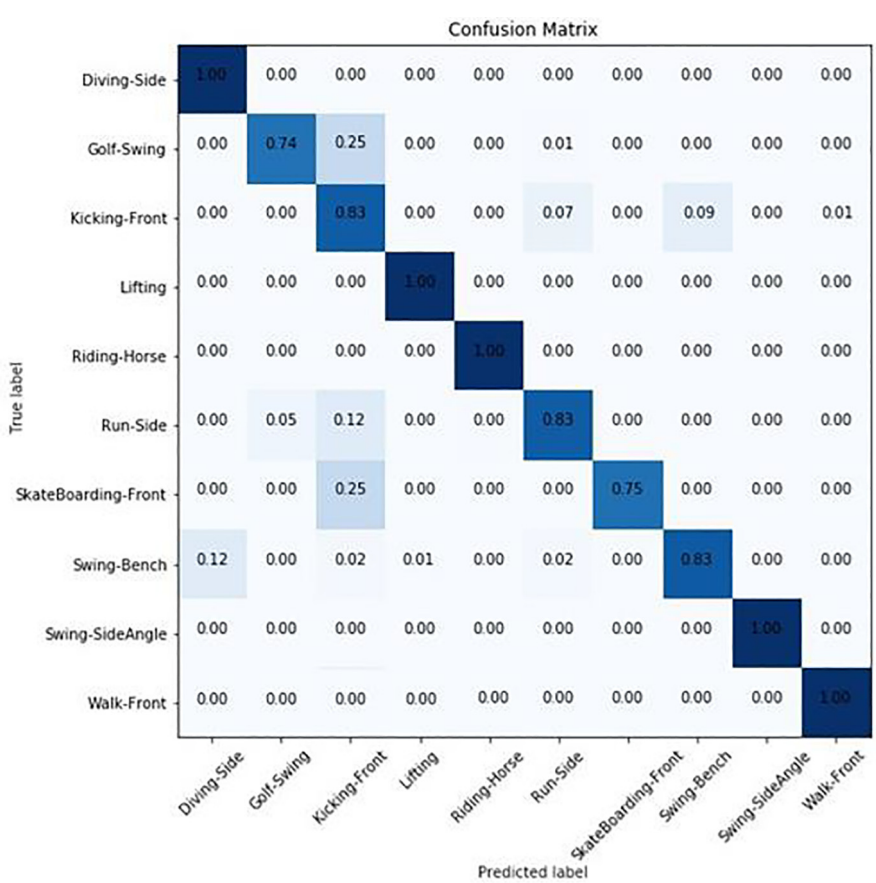

Fig. 3. The confusion matrix presentation of ten action of UCF Sport dataset: Our model provides a high performance in some actions. The validation accuracy achieved 100% for fives actions.

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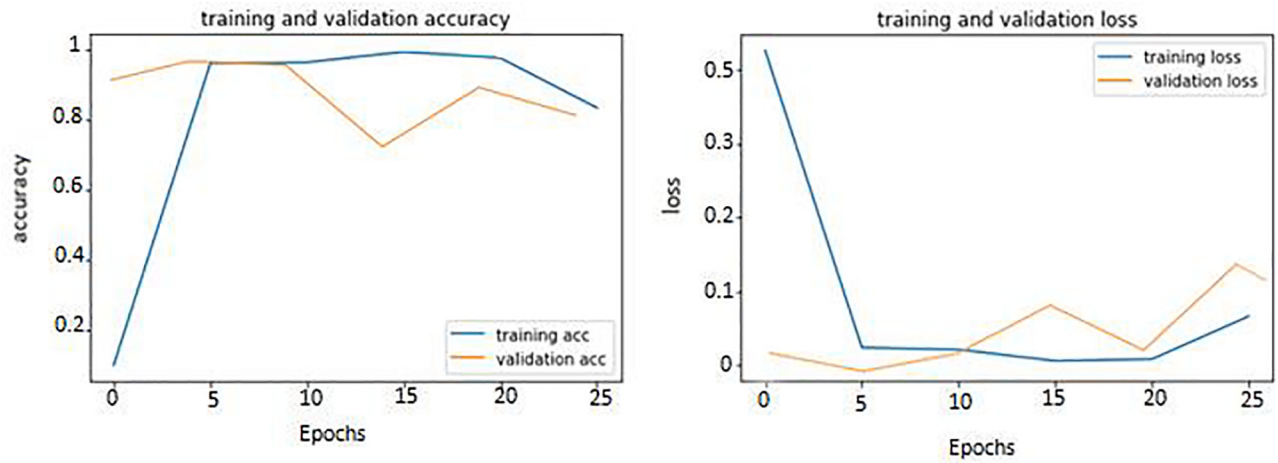

Fig. 4. The accuracy and loss presentation in the training and validation phases for UCF 101 that we used only 66 actions. These results present high validation accuracy (97%) for some actions and a weak validation around 0%.

Table 2
Performance comparison of our approach.

KTH		UCF Sport		UCF 101	
Methods	Accuracy	Methods	Accuracy	Methods	Accuracy
Rameshbabu et al., 2012	90.2%	de Oliveira Silva et al., 2017	78.46%	Soomro et al., 2012	44.5%
Latah, 2017	90.34%	Yeffet and Wolf, 2009	79.2%	Karpathy et al., 2014	65.4%
Grushin et al., 2013	90.70%	Wang et al., 2011	88%	Idrees et al., 2016	78%
Shu et al., 2014	92.30%	Our approach	89.01%	Our approach	89.30%
Geng and Song, 2015	92.49%	(GMM + KF + GRNN)		(GMM + KF + GRNN)	
Veeriah et al., 2015	93.96%				
Baccouche et al., 2011	94.39%				
Zhang et al., 2017	95%				
Shi et al., 2015	95.60%				
Our approach (GMM + KF) Jaouedi et al., 2016b	71.1%				
Our approach (GRNN)	86%				
Our approach (GMM + KF + GRNN)	96.30%				

Vector Machine to predict the human action. The accuracy rate of this model is 90.34%. Moreover our approach gives 90.01% accuracy on UCF Sport dataset compared with the proposed approach of de Oliveira Silva et al. (2017), which based on a Two-stream Convolutional Network Classifier. Here the accuracy rate does not exceed 78.46%. Furthermore, we have obtained 89.30% on UCF 101 dataset much better than Idrees et al., 2016, who got accuracy 78%.

5. Conclusion

This work presents a novel approach in the domain of human action recognition. This approach is based on the analysis of video content and extraction features. The Motion features are presented by human motion tracking using GMM and KF methods. Then others features are based on all visual characteristic of each frame on video sequence using Recurrent Neural Networks model with Gated Recurrent Unit. The main advantages of this novel approach are the analysis and the extraction of all features in each time and in each frame of video. This hybrid model has an important role to better human action recognition. The experimental results of this paper illustrate the strong performance of our novel approach to obtain high classification rate. Indeed our proposed method can be used in a diverse applications and domains like human activity analysis, medical application based on body movement, intelligent interface using hand gestures and more others. For future work, the goal is to evaluate our approach to reduce the time of video classification and to ameliorate the classification rate of UCF Sport, UCF101 and more challenging datasets.

Conflict of interest

The authors declare that there is no conflict of interest.

Acknowledgment

This work was supported and financing by the Ministry of Higher Education and Scientific Research of Tunisia.

References

Baccouche, Moez, Mamalet, Franck, Wolf, Christian, Garcia, Christophe, Baskurt, Atilla, 2011. Sequential deep learning for human action recognition. In: International Workshop on Human Behavior Understanding 2011. https://doi.org/10.1007/978-3-642-25446-8_4.

Cho, K., van Merienboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., et al., 2014. Learning phrase representations using RNN encoder–decoder for statistical machine translation. In: Proceedings of the Empirical Methods in Natural Language Processing, Oct. 25–29, Association for Computational Linguistics, Doha, Qatar, pp. 1724–1735. https://doi.org/10.3115/v1/D14-1179.

Choi, Sungjoon, Kim, Eunwoo, Songhwai, 2013. Oh human behavior prediction for smart homes using deep learning. In: IEEE RO-MAN: The 22nd IEEE International Symposium on Robot and Human Interactive Communication Gyeongju, Korea, August 26–29, 2013. https://doi.org/10.1109/roman.2013.6628440.

Chorowski, J.K., Bahdanau, D., Serdyuk, D., Cho, K., Bengio, Y., 2015. Attention-based models for speech recognition. Adv. Neural Inf. Process. Syst.

de Oliveira Silva, Vinicius, de Barros Vidal, Flavio, Romariz, Alexandre Ricardo Soares, 2017. Human action recognition based on a two-stream convolutional

network classifier. In: International Conference on Machine Learning and Applications. https://doi.org/10.1109/icmla.2017.00-64.

Deldjoo, Yashar, Quadrana, Massimo, Elahi, Mehdi, Cremonesi, Paolo, 2017. Using mise –en –scene visual features based On MPEG7 and Deep learning for movie recommendation, ArXiv, 2017, arXiv: 1704.06109.

Dey, N., Dey, N., Ashour, A., Acharjee, S., 2016. Applied Video Processing in Surveillance and Monitoring Systems. IGI Global.

Eyben, F., Wollmer, M., Schuller, B., Graves, A., 2009. From speech to letters – using a novel neural network architecture for grapheme based ASR. In: Proceedings of the IEEE Workshop on Automatic Speech Recognition and Understanding, Nov. 13-Dec. 17. IEEE Xplore Press, Merano, Italy, pp. 376–380. https://doi.org/10.1109/ASRU.2009.5373257.

Geng, Chi, Song, JianXin, 2015. Human action recognition based on convolutional neural networks with a convolutional auto-encoder. In: 5th International Conference on Computer Sciences and Automation Engineering (ICCSAE 2015) https://doi.org/10.2991/iccsae-15.2016.173.

Geng, C., Song, J.X., 2015b. Human action recognition based on convolutional neural networks with a convolutional auto-encoder. Proceedings of the 5th International Conference on Computer Sciences and Automation Engineering, (SAE' 15). Atlantis Press. https://doi.org/10.2991/iccsae-15.2016.173.

Gowsikhaa, D., Abirami, S., Baskaran, R., 2012. Automated human behavior analysis from surveillance videos: a survey. Artif. Intell. Rev. https://doi.org/10.1007/s10462-012-9341-3.

Graves, A., Jaitly, N., 2014. Towards end-to-end speech recognition with recurrent neural networks. In: Proceedings of the 31st International Conference on International Conference on Machine Learning, Jun. 21–26, JMLR.org, pp. 1764–1772.

Graves, A., Liwicki, M., Fernández, S., Bertolami, R., Bunke, H., et al., 2009. A novel connectionist system for unconstrained handwriting recognition. IEEE Trans. Patt. Anal. Mach. Intell. 31, 855–868. https://doi.org/10.1109/TPAMI.2008.137.

Grushin, A., Derek, D.M., James, A.R., Ajay Mishra, 2013. Robust human action recognition via long short-term memory. In: The 2013 International Joint Conference on Neural Networks (IJCNN), 2013, pp. 1–8.

Hochreiter, S., Schmidhuber, J., 1997. Long short-term memory. Neural Comput. 9, 1735–1780. https://doi.org/10.1162/neco.1997.9.8.1735.

Idrees, Haroon, Zamir, Amir R., Jiang, Yu-Gang, Gorbane, Alex, Laptev, Ivan, Sukthankar, Rahul, Shaha, Mubarak, 2016. The THUMOS challenge on action recognition for videos "in the wild". Comput. Vis. Image Underst. https://doi.org/10.1016/j.cviu.2016.10.018.

Jang, D.S., Jang, S.W., Choi, H.I., 2002. 2D human body tracking with structural Kalman filter. Patt. Recognit. 35, 2041–2049. https://doi.org/10.1016/S0031-3203(01)00201-1.

Jaouedi, Neziha, Zaghbani, Soumaya, Boujnah, Noureddine, Bouhlel, Mohamed Salim, 2016. Human motion detection and tracking. In: The 9th International Conference on Machine Vision 2016. https://doi.org/10.1117/12.2268539.

Jaouedi, Neziha, Boujnah, Noureddine, Htiwich, Oumayma, Bouhlel, Mohamed Salim, 2016. Human action recognition to human behavior analysis. In: The International Conference on Sciences of Electronic, Technologies of Information and Telecommunications (SETIT 2016). https://doi.org/10.1109/SETIT.2016.7939877.

Yang, Jian Bo, Nguyen, Minh Nhut, San, Phyo Phyo, Li, Xiao Li, Krishnaswamy, Shonali, 2015. Deep convolutional neural networks on multichannel time series for human activity recognition. In: Proceedings of the Twenty-Fourth International Joint Conference on Artificial Intelligence (IJCAI 2015), pp. 3995–4001.

Kale, G.V., Patil, V.H., 2016. A study of vision based human motion recognition and analysis. Int. J. Ambient Comp. Intell. (IJACI) 7 (2), 75–92.

Karpathy, A., Toderici, G., Shetty, S., Leung, T., Sukthankar, R., Fei-Fei, L., 2014. Large-scale video classification with convolutional neural networks. IEEE Conf. Computer Vision Pattern Recog. https://doi.org/10.1109/cvpr.2014.223.

Kim, H., Chalidabhongse, T.H., Harwood, D., Davis, L., 2005. Real-time foreground-background segmentation using codebook model. Real-Time Imaging 11, 172–185. https://doi.org/10.1016/j.rti.2004.12.004.

Ko, Teddy, 2011. A survey on behaviour analysis in video surveillance applications. Video Surveillance. https://doi.org/10.5772/15302.

Latah, Majd, 2017. Human action recognition using support vector machines and 3D convolutional neural networks. Int. J. Adv. Intelligent Inf. https://doi.org/10.26555/ijain.v3i1.89.

Liu, G., Tang, X., Huang, J., Liu, J., Sun, D., 2007. Hierarchical model-based human motion tracking via unscented Kalman filter. In: Proceedings of the International Conference on Computer Vision, Oct. 14–21. IEEE Xplore Press, Rio de Janeiro, Brazil, pp. 1–8. https://doi.org/10.1109/ICCV.2007.4408941.

Lopatovska, Irene, Arapakis, Ioannis, 2011. Theories, methods and current research on emotions in library and information science, information retrieval and human–computer interaction. Inf. Process. Manage. https://doi.org/10.1016/j.ipm.2010.09.001.

McFarlane, N., Schofield, C., 1995. Segmentation and tracking of pigglets in images. Mach. Vis. Appl. 8 (3), 187–193.

Patel, H.A., Thakore, D.G., 2013. Moving object tracking using Kalman filter. Int J. Comput. Sci. Mobile Comput. 2, 326–332.

Pulgarin-Giraldo, J.D., Ruales-Torres, A.A., Álvarez-Meza, A.M., Castellanos-Dominguez, G., 2017. Relevant kinematic feature selection to support human action recognition in MoCap data. In: International Work-Conference on the Interplay Between Natural and Artificial Computation. Springer, Cham, pp. 501–509.

Rameshbabu, K., Swarnadurga, J., Archana, G., Menaka, K., 2012. Target tracking system using Kalman filter. Int J. Adv. Eng. Res. Stud. 2, 90–94.

Sak, H., Senior, A., Beaufays, F., 2014. Long short-term memory recurrent neural network architectures for large scale acoustic modeling. In: 15th Annual Conference of the International Speech Communication Association, Sept. 14–18, Singapore, pp. 338–342. https://www.isca-speech.org/archive/interspeech_2014/i14_0338.html.

Shi, Yemin, Zeng, Wei, Huang, Tiejun, Wang, Yaowei, 2015. Learning deep trajectory descriptor for action recognition in videos using deep neural networks. In: IEEE International Conference on Multimedia and Expo (ICME), 2015, pp. 1–6. https://doi.org/10.1109/icme.2015.7177461.

Shu, Na, Tang, Q., Liu, Haihua, 1014. A bio-inspired approach modeling spiking neural networks of visual cortex for human action recognition. In: International Joint Conference on Neural Networks (IJCNN), p. 3450-3457.

Soomro, K., Zamir, A.-R., Shah, M., 2012. UCF101: a dataset of 101 human actions classes from videos in the wild. arXivpreprint arXiv:1212.0402.

Stauffer, C., Grimson, W.E.L., 1999. Adaptive background mixture models for real-time tracking. In: Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition, Jun. 23–25. IEEE Xplore Press, Fort Collins, CO, USA, pp. 246–252. https://doi.org/10.1109/CVPR.1999.784637.

Toshev, A., Szegedy C., 2014. Deep pose: human pose estimation via deep neural network. In: CVPR, 2014. https://doi.org/10.1109/CVPR.2014.214.

Veeriah, V., Zhuang, N., Qi, G.-J., 2015. Differential recurrent neural networks for action recognition. IEEE Int. Conf. Comp. Vision 2015, 4041–4049.

Vera, L., Gimeno, J., Coma, I., Ferńandez, M., 2011. Augmented mirror: interactive augmented reality system based on Kinect. In: Human-Computer Interaction–INTERACT 2011, pp. 483–486, 2011. https://doi.org/10.1007/978-3-642-23768-3_63.

Vinyals, O., Toshev, A., Bengio, S., Erhan, D., 2015. Show and tell: A neural image caption generator. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, Jun. 7-12. IEEE Xplore Press, Boston, MA, USA, pp. 3156–3164. https://doi.org/10.1109/CVPR.2015.7298935.

Wang, H., Klaser, A., Schmid, C., Liu, C.-L., 2011. Action recognition by dense trajectories. In: CVPR. https://doi.org/10.1109/CVPR.2011.5995407.

Wang, Heng, Schmid, Cordelia, 2013. Action recognition with improved trajectories. In: ICCV 2013 IEEE International Conference on Computer Vision, Dec 2013, Sydney, Australia. IEEE, pp. 3551–3558. <10.1109/ICCV.2013.441>. <hal-00873267v2>.

Wollmer, M., Eyben, F., Schuller, B., Rigoll, G., 2011. A multi-stream ASR framework for BLSTM modeling of conversational speech. In: Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing. IEEE Xplore Press, Prague, Czech Republic, pp. 4860–4863. https://doi.org/10.1109/ICASSP.2011.5947444.

Wren, C., Azarbayejani, A., Darrell, T., Pentland, A., 1997. Pfinder: Real-time tracking of the human body. IEEE Trans. Patt. Anal. Mach. Intell. 19, 780–785. https://doi.org/10.1109/34.598236.

Xu, W., Yang, M., Yu, K., 2010. 3D convolutional neural networks for human action recognition. In: Proceedings of the 27th International Conference on Machine Learning, Jun. 21â€"24, Haifa, Israel, pp. 495–502. https://doi.org/10.1109/TPAMI.2012.59.

Yeffet, L., Wolf, L., 2009. Local trinary patterns for human action recognition. In: ICCV. https://doi.org/10.1109/ICCV.2009.5459201.

Yi Jin, L., Abdullah, O., Shahrul, N.-S., et al., 2016. Factors influenc-ing online shopping behavior: the mediating role of purchase intention. Procedia Economics Finance. https://doi.org/10.1016/S2212-5671(16) 00050-2.

Zhang, Ning, Hu, Zeyuan, Lee, Suk-Hwan, Lee, Eung-Joo, 2017. Human action recognition based on globalsilhouette and local optical flow, In: 2nd International Symposium on Mechanical Engineering and Material Science. https://doi.org/10.2991/ismems-17.2018.1.